Local AI with Ollama - Lab Report

Q1. What is Local AI?

Local AI means running an AI model on the computer or runtime that hosts it, instead of sending each prompt to an external AI service.

Q2. Cloud LLM vs Local LLM

A Cloud LLM runs on a provider's remote servers and needs an Internet connection. A Local LLM runs where its model is installed, giving the user more control over data and offline use after setup.

Q3. Role of Ollama

Ollama downloads, manages, and runs local models. It also provides a local HTTP API that applications can use to send prompts and receive answers.

Q4. Model quantization

Quantization stores model weights using fewer bits. It reduces model size and memory requirements, often with a small possible reduction in answer quality.

Q5. Model size and memory

Larger models contain more parameters and need more memory to store their weights. They may also need additional memory for context and intermediate calculations.

Q6. Why choose a smaller model?

A smaller model can be faster, use less memory, and run on less powerful hardware. It is a good choice when its answer quality is sufficient for the task.

Q7. Advantages of local inference

Local inference can improve privacy and control, work without an Internet connection after setup, and avoid per-token API charges.

Q8. Disadvantages of local inference

Local inference requires suitable hardware and storage. Models may be slower or less capable than cloud services, and the user must install and maintain the software.

Q9. Python application architecture

The browser sends a question to Flask. Flask manages conversation history, retrieves relevant local documents, and sends the question and context to Ollama. It returns the answer, sources, and timing information to the web page.

Q10. Role of the REST API

A REST API lets programs communicate through HTTP requests and structured data such as JSON. Python uses it to send prompts to Ollama and receive model responses.

Q11. What is RAG?

RAG retrieves relevant passages from local documents and adds them to the prompt before the LLM answers. This grounds answers in the supplied material and can provide sources or indicate when information is unavailable.

Q12. Text LLM vs multimodal LLM

A text LLM processes text. A multimodal LLM can also process other inputs, such as images. For an image, it receives visual information encoded from the image pixels.

Q13. Experimental model comparison

qwen3.5: 6.55 GB on disk, 5.61 GB loaded, 27.21 s load time, 51.87 s response time, 749 tokens, 30.84 tokens/s, and 20.7% system RAM. gemma3:latest: 3.35 GB on disk, 2.88 GB loaded, 12.96 s load time, 38.76 s response time, 784 tokens, 58.79 tokens/s, and 25.4% system RAM. In this run, Gemma was smaller and faster. Results depend on hardware and settings.

Q14. Model for limited RAM

I would deploy gemma3:latest on a computer with limited RAM because it used less loaded memory in my measurements (2.88 GB versus 5.61 GB). I would first check that its answer quality is sufficient for the application.

Q15. Security risks

If Ollama is exposed to a network without authentication, others may use its compute or access prompts and data. Malicious documents may also contain prompt-injection instructions. I would restrict access with a firewall, keep the service bound to localhost when possible, and use authentication and TLS through a properly configured reverse proxy if remote access is needed.